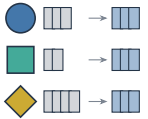
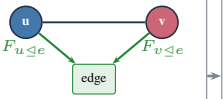
1 Encode nodes



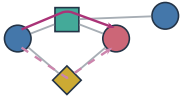
per-type encoder $\rightarrow h_v^{(0)}$
+ neighbour summary c_v
(mean per relation)

2 Edge maps



messages are compared
after the edge maps:
neighbours need not agree
directly (heterophily)

3 Sample instances



random walks on the
original graph

no meta-path list;
unbiased weight m_f

4 Weight instances



$\omega_f = m_f \ \exp g(\cdot)$

from the instance's own
nodes and relation sequence

one weight per instance

5 Instance terms



z_f = average of $T_v x_v$

T_v: edge maps composed
along the path

every node, incl. A7,
is corrected toward z_f

6 Operator & filter

$\hat{\Delta}$ = normalized $(\Delta_0 + L_P)$
edge terms + instance terms

$A = G\hat{\Delta}B$
listen / broadcast gates

$x = \sum_k T_k(A-I)\, h^{(0)} W_k$
Chebyshev filter bank

$\hat{y} = \mathrm{MLP}([x, h^{(0)}])$